

INSTAGRAM REELS VIRALITY ANALYSIS

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Business Context

Instagram Reels has become one of the most crowded content formats on the platform, with creators publishing millions of short videos daily. Yet virality doesn't scale with volume, as only a handful of reels break through with real reach, shares, and engagement, while most never get past their existing follower base.

This caused the attention of Instagram's Growth and Creator Success team - that creators with nearly identical follower counts can see wildly different outcomes. That points to something other than audience size driving virality; it's likely tied to the content itself and how it's distributed.

Problem Statement

Why Are Some Reels Going Viral While Most Stay Invisible?

Objective

The objective of this project is to analyze the factors that drive Instagram reel virality by examining content, engagement, audience behaviour & distribution metrics.

The analysis seeks identify the characteristic that distinguish high – performing reels from low-performing reels and generate data-driven recommendations that can help creators maximize the reach , engagement and viral potential.

- How strong the content and opening hook are
- Whether creators are using trending audio effectively
- How viewers behave once they start watching (retention)
- Algorithmic signals like explore page boost and reach among non-followers

Dataset and Metric Overview

Dataset	100000 Instagram reels
Total metric	26
Primary metric	Virality Score (range: 0 – 100)
Stakeholders	Content Creators and Instagram Creator Insight team

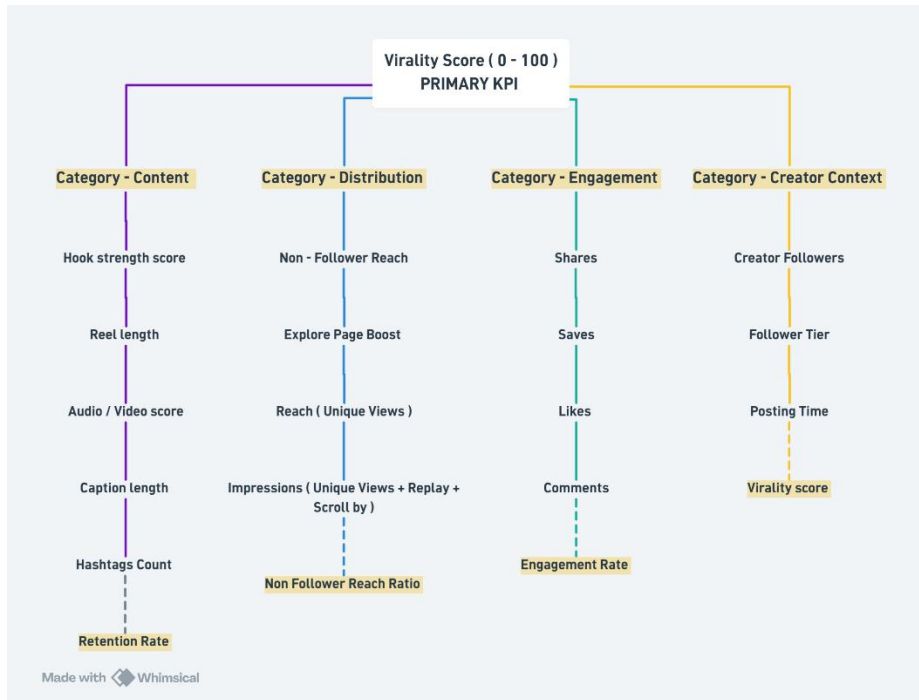
Data quality and Understanding of the data

Data quality check is performed on all the columns.

- There are no missing (null) data across all 26 metrics.
- 100000 Instagram reels are identified based on reel_id and is a unique identifier. There are no duplicates of reel_id and are unique.
- There are no duplicates of creator_id and are unique.
- Primary metric, Virality score ranges from 0 – 100 and data (all reels) is in the bounded range.

- All data in metrics are within documented bound range (0-1). (Metrics checked: posting_time, hook_strength_score, audio_popularity_score, video_quality_score, editing_quality_score, retention_rate, non_follower_reach_ratio)

Virality Score Decomposition



To perform the EDA, the metrics are organized into 5 categories aligned with the problem statement.

Category Analysis	Metrics	Metrics Count
Content Format	1. hook_strength_score 2. reel_length_sec 3. video_quality_score 4. editing_quality_score 5. caption_length 6. hashtags_count	6
Audio Strategy	1. trending_audio 2. audio_popularity_score	2
Watch Behaviour	1. avg_watch_time_sec 2. retention_rate 3. completion_rate 4. rewatch_rate 5. saves 6. likes 7. comments 8. shares	8

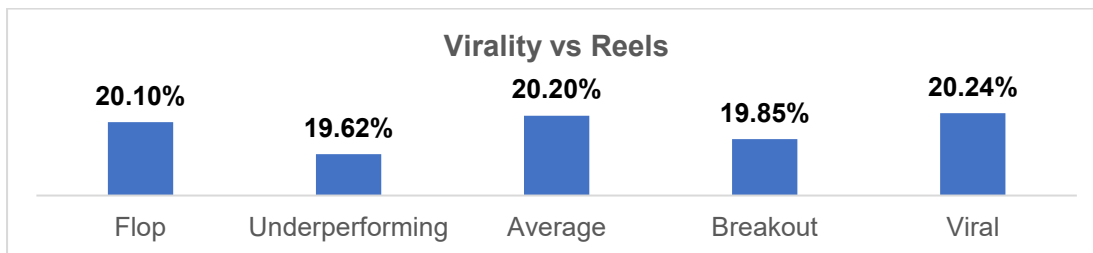
Algorithmic Distribution	1. explore_page_boost 2. non_follower_reach_ratio 3. impressions 4. reach 5. engagement_rate	5
Creator & Timing	1. creator_followers 2. follower_tier 3. posting_time	3

Derived Tiers added

Virality Tier Bucketing Logic

Purpose - To classify reels into performance tiers and identify the factors driving virality.

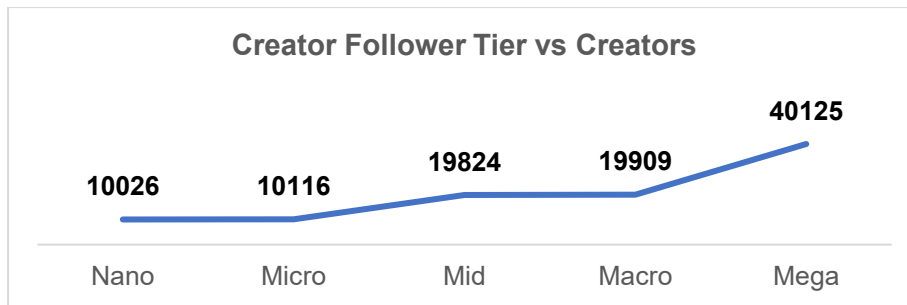
Virality Score range	Tier	Count of reel_id	% of Total reels
0-20	Flop	20102	20.10%
20-40	Underperforming	19621	19.62%
40-60	Average	20196	20.20%
60-80	Breakout	19845	19.85%
80-100	Viral	20236	20.24%
	Grand Total	100000	100.00%



- Roughly 20% of reels fall in each tier, with an even spread across the performance spectrum.

Creator follower Tier Bucketing Logic

Creator follower range	Tier	creator_id count	% of Total reels
<1M	Nano	10026	10.03%
1M-2M	Micro	10116	10.12%
2M-3M	Mid	19824	19.82%
3M-4M	Macro	19909	19.91%
4M-5M	Mega	40125	40.13%
	Grand Total	100000	100.00%



- 40.1% of Creators have 4M-5M followers, ~40% of Creators have 2M-4M followers and only ~20% creators have <2M followers.

Hypothesis and Analysis

Hypothesis Testing

1. High virality reels generate high impressions.
2. Higher saves lead to higher virality score.
3. Higher likes lead to higher virality score.
4. Posting at certain hours drives higher virality.
5. More hashtags show diminishing returns on virality beyond a threshold.

Correlation between Virality and Other Metrics

Correlation between Virality and Metrics			
S.No	Metrics	Correlation	ABS Correlation
1	shares	0.00558	0.0056
2	non_follower_reach_ratio	0.00478	0.0048
3	posting_time	-0.00475	0.0048
4	creator_followers	-0.00413	0.0041
5	saves	0.00410	0.0041
6	hashtags_count	-0.00401	0.0040
7	engagement_rate	-0.00339	0.0034
8	explore_page_boost	0.00319	0.0032
9	caption_length	0.00293	0.0029
10	hook_strength_score	-0.00275	0.0028
11	completion_rate	-0.00265	0.0027
12	reach	-0.00225	0.0023
13	avg_watch_time_sec	-0.00221	0.0022
14	likes	-0.00159	0.0016
15	retention_rate	-0.00138	0.0014
16	audio_popularity_score	-0.00137	0.0014
17	impressions	0.00135	0.0014
18	comments	-0.00132	0.0013
19	video_quality_score	-0.00125	0.0012
20	trending_audio	-0.00103	0.0010
21	reel_length_sec	-0.00055	0.0005
22	editing_quality_score	0.00041	0.0004

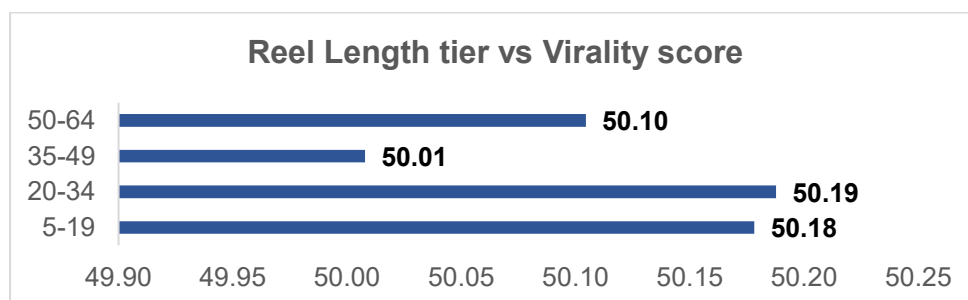
23	rewatch_rate	0.00001	0.0000
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Insights

- Shares have the highest correlation and is positive ($r=+0.0056$) with virality score of any metric in the dataset.
- Non-follower reach & saves are next ($r=+0.0048$, $+0.0041$) highest positive correlation.
- Posting time & creator followers have highest negative correlation ($r=-0.0048$, -0.0041).
- Rewatch rate has ~ 0 correlation with virality score.

Category 1: Content Format Analysis

Content metrics by Virality Tier					
Virality Tier	Avg caption length	Avg Hook Strength (0 - 1)	Avg hashtags count	Avg Editing Quality Score (0 - 1)	Avg Video Quality Score (0 - 1)
Flop	149.73	0.4997	15.07	0.4991	0.4995
Underperforming	149.38	0.5004	15.21	0.4983	0.5019
Average	149.82	0.4970	15.03	0.4993	0.4974
Breakout	150.26	0.5002	15.01	0.5002	0.4984
Viral	150.19	0.4978	15.03	0.4995	0.5010



Insights

- The higher avg caption length from Breakout to Viral Tiers shows a possibility of well written content for high viral videos. Therefore, caption length with **150+ characters** pulls virality.
- Avg hashtags count are higher in Flop and Underperforming tier. Usage of less hashtags in reels helps in virality. Therefore, the usage of relevant hashtags could matter more than n. of hashtags.
- Breakout and Viral tier have higher avg Editing quality score than any other virality tier, which implies higher the editing quality score - higher is virality and is a strong point for the creators.
- The reel length has been categorized into 15-second groups, Group 5-19 and 20-34 have higher avg virality score. So avg reel duration of less than **35 seconds** shows a higher virality score avg, implying that short videos grab more attention.

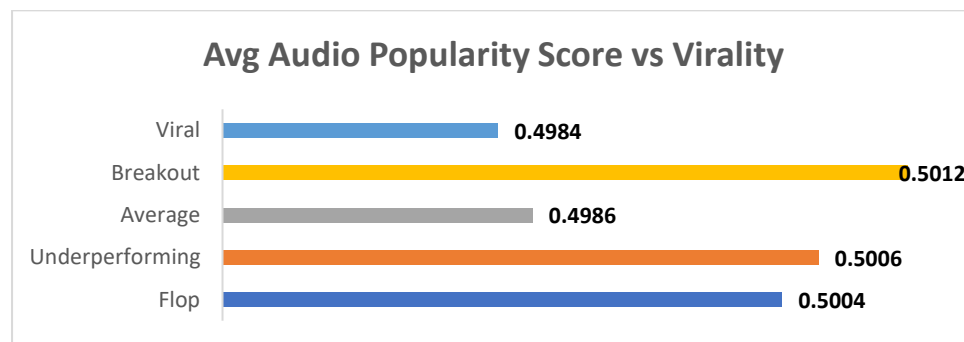
Hypothesis 5 is not supported: More hashtags show diminishing returns on virality beyond a threshold.

Avg Hashtag count doesn't show a clear threshold effect across the tiers. There is a avg ~15 hashtag count across tiers and is negatively correlated, where $r=-0.0040$.

Category 2: Audio Strategy Analysis

Audio popularity metrics by Virality Tier			
Virality tier	Count of reel_id	% of trending audio	Avg Audio Popularity Score (0 - 1)
Flop	20102	50.44%	0.5004
Underperforming	19621	49.56%	0.5006
Average	20196	49.42%	0.4986
Breakout	19845	50.04%	0.5012
Viral	20236	50.12%	0.4984
Grand Total	100000	49.92%	0.4999

Trending Audio: Direct Comparison (Yes vs No)					
Trending audio	Count of reel_id	Avg virality_score	Avg engagement_rate	Avg non_follower_reach_ratio	Avg audio_popularity_score
FALSE	50083	50.1502	0.5030	0.5002	0.4992
TRUE	49917	50.0907	0.4995	0.5005	0.5005
Grand Total	100000	50.12	0.5012	0.5004	0.4999



Insights

- Audio popularity score averages 0.498 - 0.501 across all five tiers. High-popularity audio is equally present in Flop reels as well as Viral ones.
- Trending audio is used by ~50% of creators across every virality tier with no clustering at the Viral end.
- The trending-audio True/False comparison show non-follower reach ratio at 0.5005/ 0.5002 (minor difference) and engagement rate at (0.4995 / 0.5030).
- No evidence that trending audio drives the algorithm to distribute beyond the creator's existing audience. **Therefore, Audio strategy has zero virality lift and no differentiator.**

Category 3: Watch Behaviour Analysis

Watch behaviour by Virality Tier				
Virality tier	Average watch time	Avg Completion Rate (0 - 1)	Avg Retention Rate (0 - 1)	Avg Rewatch Rate (0 - 1)
Flop	29.98	0.5013	0.5035	0.4972
Underperforming	30.27	0.5030	0.4989	0.5039
Average	30.07	0.4997	0.5010	0.4990
Breakout	29.87	0.4987	0.4989	0.4944
Viral	30.03	0.5006	0.5025	0.5002
Grand Total	30.04	0.5007	0.5010	0.4989

Insights

- Flop and Underperforming tiers have higher avg completion rate, Retention rate, rewatch rate and Avg watch time.
- Completion rate and Retention rate, both show less than 0.5% variation across all virality tiers (Flop to Viral).
- Higher Avg Rewatch rates are present in both top & bottom tiers (Underperforming and Viral).

Therefore, watch-through behaviour does not emerge as a linear driver of virality.

Virality Tier V/S Audience Behaviour				
Virality tier	Avg saves	Avg likes	Avg comments	Avg shares
Flop	14929.32	501182.54	25046.22	14887.75
Underperforming	14985.30	499876.69	25005.09	14922.75
Average	14977.11	497229.49	25077.33	15023.87
Breakout	15075.16	499177.07	24940.94	15105.30
Viral	15013.82	499455.44	25072.91	14973.52
Grand Total	14996.00	499380.48	25028.94	14982.64

Insights

- Breakout and Viral tiers have higher average saves.
- Flop and Underperforming tiers have higher average likes.
- Avg comments have spread and lower significant difference across virality tiers.
- Avg shares are higher in tiers above the Average tier. And shares have the highest correlation ($r=+0.0056$)

Therefore, Audience behaviour has significant drivers in the virality. Higher the shares and saves, higher is the virality. A high number of likes has low virality.

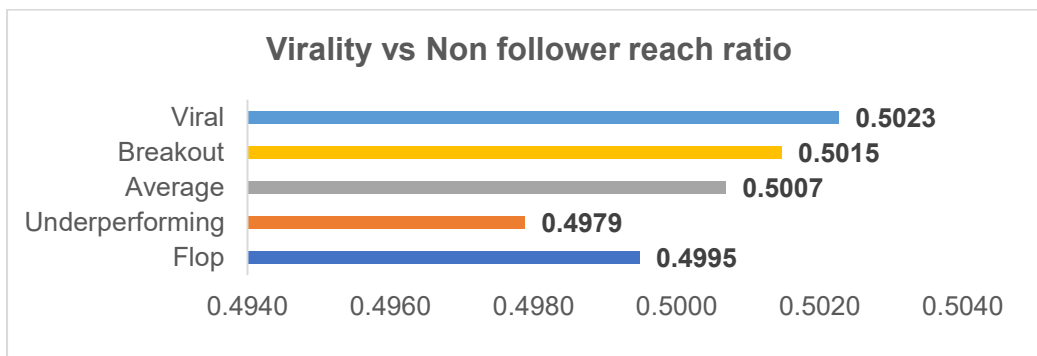
Hypothesis 2 supported: Higher shares lead to higher virality score and has strong positive correlation of $+0.0056$.

Hypothesis 3 not supported: Higher likes do not lead to higher virality score and are negatively correlated ($r=-0.0016$)

Category 4: Algorithmic Distribution Analysis

Distribution metrics by Virality Tier

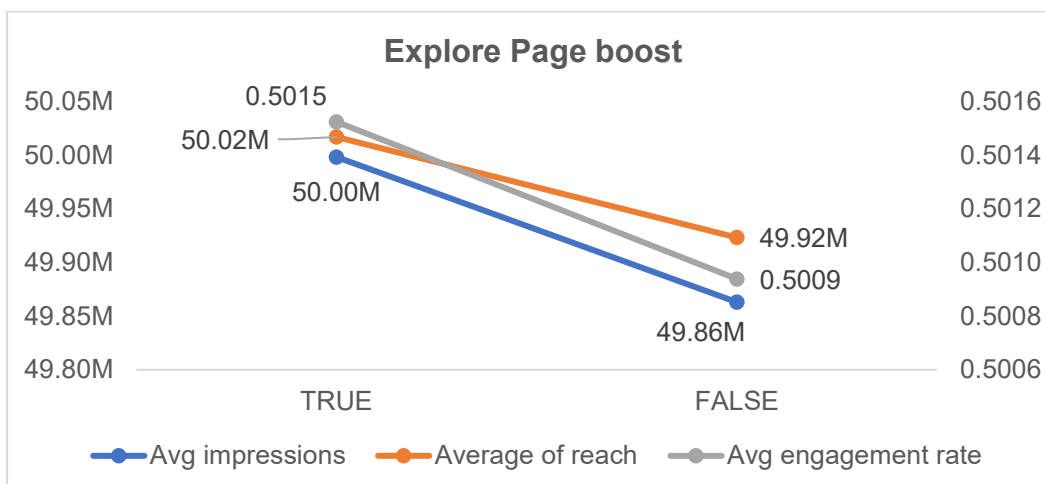
Virality tier	Avg impressions	Avg reach	Avg engagement_rate	Avg Non follower reach ratio (0 - 1)	Avg explore_page_boost2
Flop	50.07M	49.97M	0.5013	0.4995	0.4976
Underperforming	49.83M	50.18M	0.5010	0.4979	0.4957
Average	49.63M	50.17M	0.5056	0.5007	0.4924
Breakout	50.07M	49.62M	0.4989	0.5015	0.4954
Viral	50.05M	49.90M	0.4993	0.5023	0.5030
Grand Total	49.93M	49.97M	0.5012	0.5004	0.4968



Insights

- Non-follower reach ratio: Non-follower reach ratio ranges from 0.497 (Underperforming) to 0.502 (Viral). While the gap is narrow, the direction is consistent and shows a linear growth with respect to virality tier.
- Higher non-follower reach could mean a higher virality score.

Hypothesis 1 not supported: Impressions are virtually identical across all virality tiers (~50M avg). No meaningful separation and Correlation is positive ($r=+0.0014$), but the analysis did not result in a positive relationship between virality and total impressions.

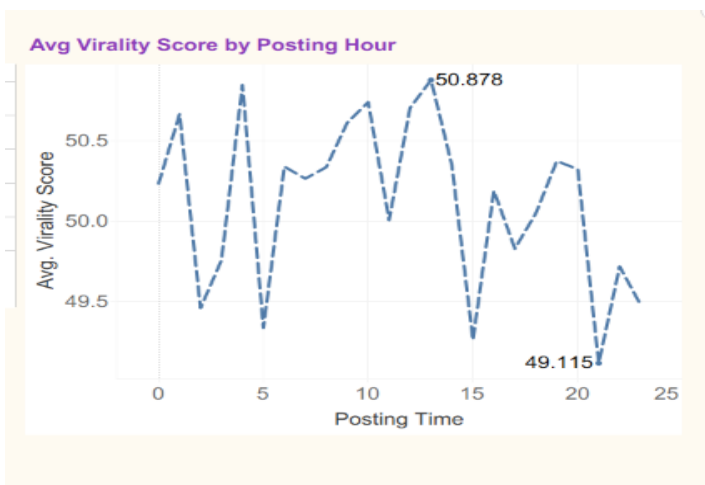


Insights

- Explore-boosted reels average virality 50.21 vs 50.03 for non-boosted - a 0.18 gap. It shows a positive but modest relationship with virality and also positively correlated ($r=+0.0032$)
- Explore-boosted reels have higher reach, avg impressions, avg engagement rate. Creator reels using explore page boost have greater inorganic growth (impression->reach->engagement rate)

Category 5: Creator and Timing Analysis

Virality Score by Follower Tier (Nano through Mega)		
Creator - Follower ti	Average of creator_followers	Avg virality_score
1.Nano	253841.25	50.5437
2.Micro	748210.30	50.3085
3.Mid	1498689.04	50.0829
4.Macro	2496845.23	50.0382
5.Mega	4001838.54	50.0268
Grand Total	2501073.82	50.1205



Insights

- Nano creators average virality 50.54 vs Mega (3M–5M) at 50.03. Nano being higher and outperforming. This clarifies Growth teams statement, that follower count doesn't determine the virality.
- Hour 13 (1pm) has the at 50.88 avg virality being highest, followed by Hour 4 (4am) and Hour 1 (1 am) at 50.85 and 50.67.
- Lowest performing: Hour 21 (9pm) at 49.12, Hour 5 (5am) at 49.33, Hour 15 (3pm) at 49.26.
- Top-to-bottom spread: 1.76 virality points, the widest variation variable in the dataset.
- Posting time is the most differentiated variable we have, suggesting timing does play a role in content distribution.

Hypothesis 4 is supported: Posting at certain hours drives higher virality.

Hour 13 , 4 , 1 have shown a comparatively higher avg virality score (50.88 – 50.67). Posting at this hour can help with reel going viral.

Hypothesis Testing Results

Hypothesis	Result	Correlation (r)
1. High-virality reels generate high impressions.	Rejected	0.0014
2.Higher shares lead to a higher virality score.	Accepted	0.0056
3.Higher likes lead to a higher virality score.	Rejected	-0.0016
4.Posting at certain hours drives higher virality.	Accepted	-0.0048
5.More hashtags show diminishing returns on virality beyond a threshold.	Rejected	-0.004

Recommendations

1. Prioritize shares and saves over likes as core engagement KPIs.

Shares showed the highest positive correlation with virality ($r = +0.0056$), followed by saves ($r = +0.0041$), while likes were negatively correlated ($r = -0.0016$). Creator guidance and in app prompts should be weighted toward save and share worthy content rather than like driven content.

2. Optimize for Explore page eligibility and non-follower reach.

Explore boosted reels averaged a 0.18 point higher virality score than non-boosted reels (50.21 vs 50.03, $r = +0.0032$), and non-follower reach ratio rose consistently from Underperforming to Viral tiers. Distribution mechanics should be treated as a lever, not just a byproduct of content quality.

3. Standardize posting around Hour 13 (1pm), Hour 4 (4am), and Hour 1 (1am).

These hours showed the highest average virality scores (50.88, 50.85, 50.67 respectively), while Hour 21 (9pm), Hour 5 (5am), and Hour 15 (3pm) were the lowest performers. This was the most differentiated variable in the dataset (1.76-point spread, Highest to lowest) and Hypothesis 4 was supported.

4. Set caption length guidance at 150+ characters.

Average caption length increased from Breakout to Viral tiers, suggesting more developed captions correlate with better performance.

5. Recommend fewer, more relevant hashtags rather than a higher count.

Flop and Underperforming tiers used more hashtags on average, and hashtag count was negatively correlated with virality ($r = -0.0040$), with no threshold effect observed

(Hypothesis 5 not supported). This points to hashtag relevance mattering more than volume.

6. **Invest in editing quality as a content side lever.**

Breakout and Viral tiers had the highest average editing quality scores of any tier, making this a strong, controllable factor for creators.

7. **Favor reel lengths under 35 seconds.**

Duration groups of 5-19 and 20-34 seconds had the highest average virality scores.

8. **Deprioritize trending-audio strategy.**

Audio popularity score and trending audio usage were flat across all tiers (0.498–0.501), and the True/False comparison on non follower reach and engagement rate showed negligible difference. The analysis found no evidence trending audio expands distribution.

9. **Do not treat watch through metrics (completion rate, retention rate, rewatch rate) as virality signals.**

These showed less than 0.5% variation across all tiers and were, in some cases, higher in Flop/Underperforming tiers than in Viral ones.

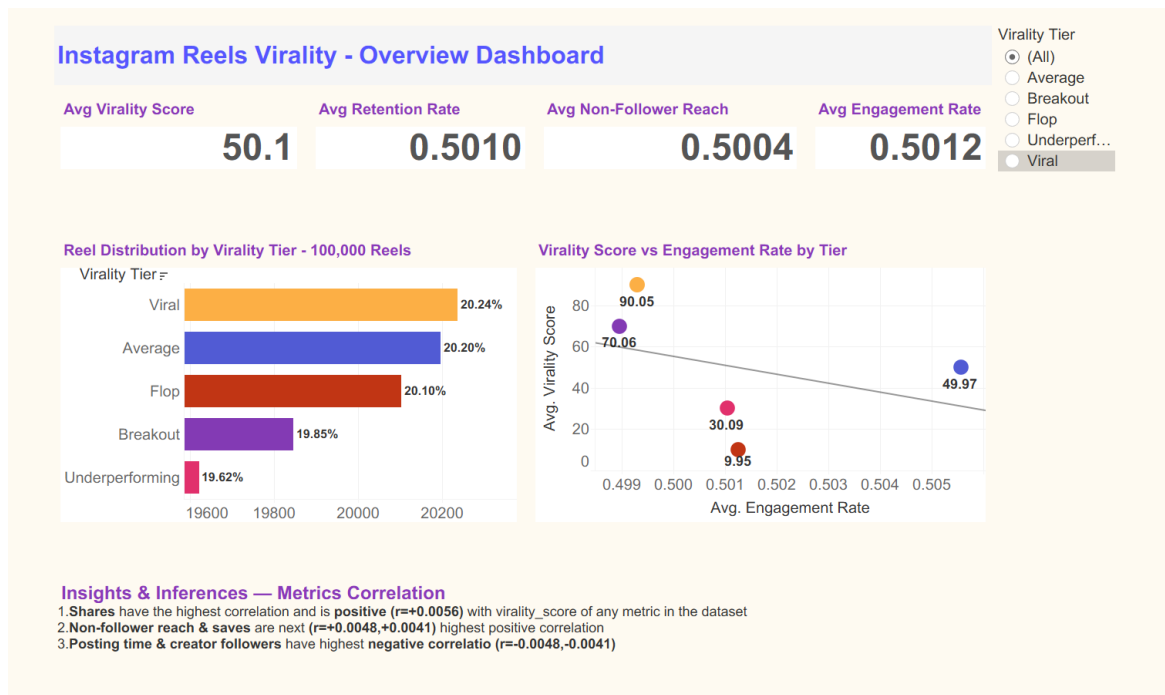
Conclusion

The analysis indicates that virality on Instagram Reels is driven primarily by audience behavior (shares, saves) and distribution mechanics (Explore page boost, non-follower reach, posting time), rather than by follower count, audio strategy, or depth of viewer watch-through. Shares and saves were the strongest positively correlated metrics in the dataset, and posting time produced the widest performance spread of any variable tested. Content side factors: editing quality, caption length, and hashtag relevance also showed consistent directional patterns, indicating that content craft still plays a supporting role.

Conversely, several commonly assumed drivers did not hold up: trending audio showed no measurable lift, impressions were not meaningfully linked to virality, and watch-through behavior did not distinguish high performers from low performers. Of the 5 hypotheses tested, two were supported (shares, posting time) and three were not (impressions, likes, hashtag threshold effect).

These findings suggest Instagram's creator strategy and platform recommendations should shift focus toward **distribution amplification and share/save-optimized content**, while deprioritizing audio-trend chasing and follower-count-based targeting as virality signals.

Tableau Dashboard overview

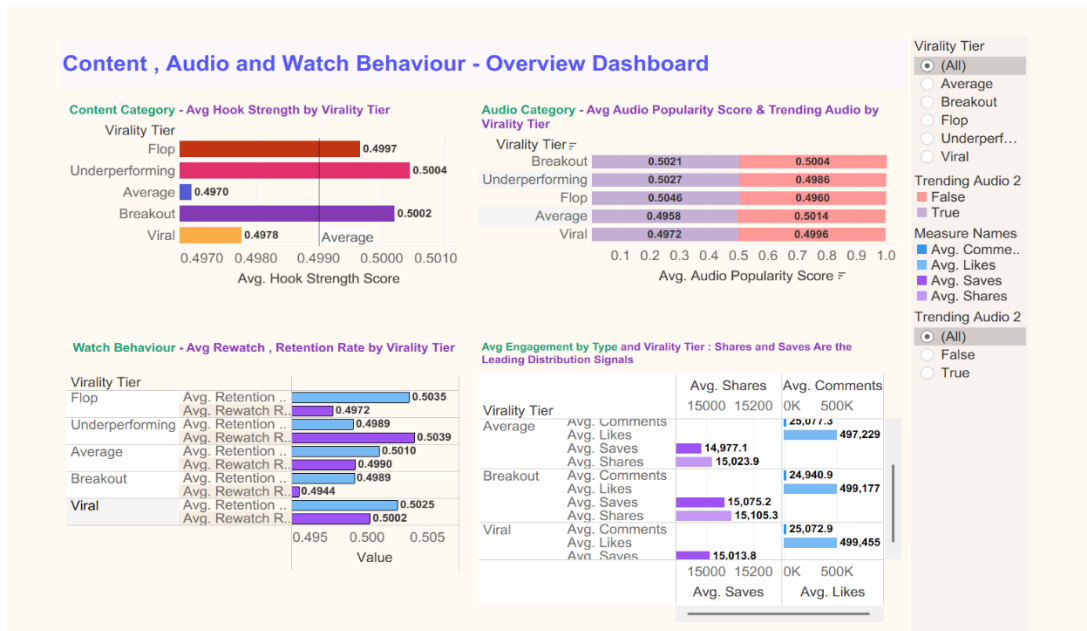


Dashboard 1 Summary -

Highlights the KPI Metric Overview & Metrics Correlation.

Chart 1 – Displays the reel distribution by virality tier. 1,00,000 reels split ~20% across all the tiers.

Chart 2 – It shows the Avg engagement rate with Avg virality scores. Viral and Breakout tiers have the lower engagement rate, with Average tier having higher engagement rate. This doesn't align with the virality.



Dashboard – 2 Summary

Chart 1 – Content category - The higher average hook strength is present in both breakout and underperforming tiers. Therefore, hook strength is not linear with virality tiers & doesn't define the virality.

Chart 2 – Audio Category – Trending Audio - 'True' – The average audio popularity scores are higher in Average , Breakout & Viral Tiers.

Chart 3 – Higher average rewatch and retention rate are present in both high viral and underperforming tiers. Hence, these are not the drivers for virality.

Chart 4 – Audience Behaviour – Shares and Saves are higher among viral tiers and they are significant drivers of virality. These also have a higher correlation with virality.



Dashboard – 3 Summary

Chart 1 – Below Average Tiers have higher engagement rate, which doesn't align with the virality tier.

Non-follower reach increases with above-average virality tiers.

Chart 2 – The chart displays Reel count by 'Follower count tier' and 'Underperforming & Flop Virality Tier '. The creators with higher follower count have higher flop & underperforming reel counts.

Chart 3 – The chart explains the Avg virality score by posting hour, posting time 13 (1pm) has the highest avg virality score and the posting time 21 (9pm) has the lowest virality score.